Maggie Zhang

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Education

University of California, San Diego — La Jolla, CA
BS Data Science, BS Cognitive Science (Machine Learning & Neural Computation) — September 2023 – March 2027

- GPA: 3.8, Minor in Linguistics

Experience

AI Brain Programmer — July 2025 – September 2025
Swartz Center for Computational Neuroscience — La Jolla, CA

- Curated and validated large-scale EEG datasets totaling 100 GB using **Hierarchical Event Descriptors (HED)** within the **BIDS** standard, producing machine-readable, analysis-ready data for cross-study reuse.
- Expanded key machine learning, visualization, and time-series analysis features in **EEGLAB**, the largest EEG ecosystem (15000+ active users), with automated testing, API documentation, and Git version control.

Data Science Intern — June 2025 – August 2025
Cenergy Power — Aliso Viejo, CA

- Built an **NLP sentiment analysis pipeline** to extract regulatory risk signals from unstructured municipal board meeting minutes, encoding **Authority-Having-Jurisdiction sentiment** toward solar permitting.
- Combined NLP-derived policy signals with **computer vision** (OpenCV, CNNs) and geospatial data into a **site-scoring algorithm**, identifying **5+ high-potential sites** and improving pipeline efficiency by **20%**.

Bioinformatics Data Intern — June 2024 – September 2024
SciCrunch — La Jolla, CA

- Improved **Python regex scripts** and **SQL** database storage to increase accuracy and efficiency of text extraction of research resource identifiers (RRIDs), with a streamlined validation pipeline for thousands of resources.
- Developed **Google Apps Script automation** to streamline data processing, optimize file management, and validate **bioinformatics** resource extraction from **100,000+ scientific articles**.

Projects

Novel Predictions: Book Recommendation | *Python, XGBoost, Surprise* — November 2025 – Present

- Built a hybrid recommendation system using **XGBoost** with an **RMSE of 0.8019** by engineering implicit user-behavior features, outperforming collaborative filtering baselines **(SVD++, RMSE 0.87)**.
- Designed an evaluation pipeline comparing gradient-boosted trees and matrix factorization models, demonstrating improved handling of **nonlinear interactions** in sparse user–item data.

Char RNN Text Modeling | *PyTorch, NLTK, Transformers, pandas* — January 2025 – March 2025

- Implemented **GRU and LSTM** character-level language models in **PyTorch** to generate stylistic text, benchmarking against an **n-gram baseline**.
- Conducted structured **hyperparameter tuning** across model depth, hidden size, and learning rate to minimize validation loss and perplexity.

Understanding Hurricanes | *D3, JavaScript, HTML, CSS, pyproj* — November 2025 – Present

- Built an **interactive data visualization platform** using **D3.js** to explore **100+ years of hurricane data**, enabling analysis of intensity, frequency, and economic impact trends.
- Designed **dynamic dashboards** with **filtering, tooltips, and simulations** to support exploratory analysis of storm destructiveness, including a focused Hurricane Sandy case study.

Technical Skills

Languages: Python, SQL, JavaScript (ES6+), Bash/Shell, MATLAB, Java, R, SAS
ML & Stats: Gradient-Boosted Decision Trees, Neural Networks, K-Means Clustering, A/B Testing, Causal Inference
NLP: TF-IDF, Topic Modeling, word2Vec, BERT, NLTK, Hugging Face Transformers, LLM-based text extraction
Data Engineering & MLOps: PySpark, Dask, AWS (EC2, S3), Databricks, Git, CI/CD, Linux/Unix
Libraries & Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face, Pandas, NumPy, Surprise
Visualization & Reporting: D3.js, Tableau, Plotly, Matplotlib, Seaborn, Streamlit, HTML/CSS

Relevant Coursework

Statistical NLP, Recommender Systems & Web Mining, Supervised & Unsupervised Machine Learning, Neural Networks, Deep Learning, Causality for ML/AI, Scalable Analytics Systems, Data Engineering, Data Analysis & Inference, Practice of Data Science, Probability, Statistics